Multi-Attribute Fair Vector Search

Technical Report: Dataset Descriptions, and Extended Related Work

1 Dataset Descriptions

We use four real-world datasets and two synthetic datasets that span diverse application domains and fairness-relevant characteristics. The real-world datasets include FairFace [40] and CelebA [50] for image embeddings, Audio [2] for speech representations, and GloVe [32] for text embeddings. We also construct two synthetic datasets to demonstrate the scalability of our frameworks in working with large-scale real-world data.

Audio. The Audio dataset is derived from the TIMIT corpus [2], a speech dataset of 53,387 utterances. Each utterance is represented as a 192-dimensional embedding capturing speaker-level acoustic characteristics. The three protected attributes are *audio quality* (512 kbps, 768 kbps, 1024 kbps), *speaker gender* (male, female), and *file format* (MP3, WAV). The dataset uses ℓ_2 distance and exhibits moderate imbalance across the quality and format groups.

FairFace. FairFace [40] is a face image dataset designed for balanced demographic representation, comprising 97,698 images. Each image is encoded as a 768-dimensional embedding using a pre-trained vision transformer. The three protected attributes are *race* (7 categories: White, Black, Latino/Hispanic, East Asian, Southeast Asian, Indian, Middle Eastern), *gender* (male, female), and *age group* (9 bins). Unlike CelebA, FairFace is explicitly balanced across race and gender, making it suitable for evaluating fairness-aware retrieval under near-uniform group distributions.

CelebA. CelebA [50] is a large-scale face attribute dataset of 202,599 celebrity images, each encoded as a 512-dimensional embedding. The dataset provides up to 40 binary attribute annotations per image (e.g., hair color, facial features, accessories). Since our framework targets a small number of semantically meaningful protected attributes, we randomly select up to 5 of the 40 binary attributes to define protected groups, yielding experiments with $m \in \{1, 2, 3, 4, 5\}$. The selected attributes exhibit significant class imbalance, providing a realistic test of retrieval under skewed group distributions.

GloVe. The GloVe dataset [32] consists of 1,183,514 pre-trained word embeddings of dimension 100, originally trained on a Common Crawl corpus. Cosine distance is used as the similarity metric, as it is standard for comparing word vectors. Three protected attributes are constructed by assigning categorical labels to words: *theme* (sports, life, war), *date* (2020, 2021), and *area* (Asia, North America), yielding a controlled multi-attribute fairness setting over text embeddings.

SIFT1B (BIGANN Benchmark). SIFT1B, also referred to as the BIGANN benchmark [34] is a standard billion scale ANN dataset comprising $n = 10^9$ 128-dimensional SIFT descriptors extracted from natural images and evaluated under ℓ_2 distance. We use the full base set of one billion vectors.[1] Because SIFT1B provides embeddings but no protected attribute annotations, we synthesize protected attributes for fairness evaluation, following exactly the schema of Synthetic 2

[1] Official source: http://corpus-texmex.irisa.fr/ (ANN_SIFT1B / BIGANN base set).

so that the two billion-scale datasets differ only in their vectors: real SIFT descriptors versus well-separated Gaussian clusters. Specifically, each vector is annotated, independently of its embedding, with a tuple of three protected attributes A_1, A_2, A_3 whose domains have $|V_1| = 3$, $|V_2| = 4$, and $|V_3| = 3$ values, yielding $3 \times 4 \times 3 = 36$ partitions. Attribute values follow fixed, slightly imbalanced categorical distributions—A_1 at $(0.40, 0.35, 0.25)$, A_2 at $(0.30, 0.25, 0.25, 0.20)$, and A_3 at $(0.40, 0.35, 0.25)$—to mimic realistic demographic skew. Since the assignment does not depend on the embeddings, it introduces no artificial correlation between protected group membership and vector geometry; the clustered structure of Synthetic 2 is thus deliberately absent, isolating the effect of using real high-dimensional data at billion scale. The dataset is stored as memory-mapped binary files (a float32 vector file of ∼512 GB and a uint8 attribute-code file of ∼3 GB), so the full billion-vector set is never loaded into RAM.

Synthetic 1. Synthetic 1 is a controlled dataset of 10,000+ vectors in $\mathbb{R}^{128}$ designed to expose the failure modes of graph-based indexing under cluster imbalance. A reference center $\mu \in \mathbb{R}^{128}$ is fixed, and two types of points are generated: (i) a small *near cluster* of vectors sampled tightly around μ (σ_{near} small), and (ii) a large *far cluster* of vectors sampled from a shifted center $\mu + \delta$ (σ_{far} larger). Query vectors are drawn near μ, so the true nearest neighbors lie in the small near cluster while the embedding space is dominated by the larger far cluster. This construction tests whether retrieval methods can surface sparse but relevant near neighbors in the presence of a dense but distant majority cluster. Three or more protected attributes are assigned to partition the data into Cartesian subgroups.

Synthetic 2. Synthetic 2 is a billion-scale dataset ($n = 10^9$, $d = 32$) designed to evaluate preprocessing and query processing scalability. Three protected attributes are defined: A_1 with 3 values, A_2 with 4 values, and A_3 with 3 values, yielding 36 Cartesian partitions. Each partition π is assigned a distinct Gaussian cluster center $\mu_\pi \sim \mathcal{N}(0, \sigma_c^2 I_d)$ with $\sigma_c = 10$, and points within partition π are sampled as $\mathcal{N}(\mu_\pi, \sigma^2 I_d)$ with $\sigma = 1$, producing well-separated clusters. Attribute proportions are slightly imbalanced to reflect realistic demographic distributions: A_1 at (0.40, 0.35, 0.25), A_2 at (0.30, 0.25, 0.25, 0.20), and A_3 at (0.40, 0.35, 0.25). The dataset is stored as memory-mapped binary files: a float32 vector file (∼128 GB) and a uint8 attribute code file (∼3 GB), enabling the framework to operate without loading the full dataset into RAM.

2 Extended Related Work

Vector Database Systems and Indexes. Vector databases have become essential infrastructure for AI applications [24, 58]. Systems such as `Milvus` [71], `Qdrant` [61], and `SingleStore-V` [9] support hybrid similarity+filter queries, while production-scale systems such as `GaussDB-Vector` [68] and `HAKES` [26] demonstrate the growing need for such capabilities at billion scale. `HARMONY` [83] and `CoTra` [96] address distributed and RDMA-accelerated vector search for high-throughput workloads. `TigerVector` [49] supports vector search within graph databases for advanced RAG pipelines. `Wolverine` [60] enables faster deletion throughput for graph index maintenance. Current indexing methods fall into four main families [58]: LSH-based [29, 30, 16, 19], graph-based [54, 64, 59, 45], tree-based [15, 55], and inverted-file (IVF)/clustering-based indexes [17, 34, 21]. IVF clusters the vectors with k-means and scans only the lists of the `nprobe` centroids nearest to the query, optionally over compressed codes [33, 21]. Partition-based filtered indexes such as CAPS [22] subdivide these clusters by attribute values; our framework partitions by attribute combination first and accepts any index, IVF included, inside each partition. Earlier advances include quantization-integrated indexing [20], production-optimized graph search [98], vector index merging [79, 37], out-of-distribution graph search [90], triangle-inequality-based pruning [82, 66], and GPU-accelerated billion-scale

search [35].

A recent comprehensive evaluation by Azizi et al. [5] benchmarks twelve graph-based methods across datasets up to one billion vectors, identifying five design paradigms. Graph-based indexing remains the dominant paradigm at SIGMOD and VLDB 2026. Li et al. [42] design proximity graphs with faster search convergence, Qiu et al. [62] organize the graph around hemi-sphere centroids, Hua et al. [27] detect and repair locally hard regions of the graph at search time, Zhang et al. [92] adapt HNSW exploration to the data distribution, and Xu et al. [84] revisit graph-based ANN search through the lens of hubness. Zhao et al. [95] extend graph indexes to maximum inner product search via norm-adaptive partitioning, and `GEM` [69] builds a native graph index for multi-vector retrieval. On the quantization and cluster-based side, which is closest to the IVF baseline in our evaluation, `RAIRS` [87] optimizes redundant assignment and list layout for IVF, `TaCo` [78] introduces a data-adaptive subspace-collision index, `SAQ` [43] pushes vector quantization via code adjustment and dimension segmentation, `JHQ` [23] combines Johnson–Lindenstrauss projections with hierarchical quantization, and `IVF−RaBitQ` [63] realizes a GPU-native IVF index. Hardware- and storage-aware designs include `GPS` [88] and Chen et al. [10] for disk-resident indexes, `CMANNS` [28], `FlashANNS` [80], Li et al. [47] and McCoy et al. [56] for GPU-accelerated construction and search, and `DiskJoin` [11] for SSD-based similarity joins. `ANNiE` [76] learns query-cost estimates for graph-based search, and `ConANN` [25] provides conformal guarantees on ANN results. All of these methods target unconstrained or Boolean-filtered similarity search and are orthogonal to our contribution: like LSH, IVF and HNSW in Section IV-B, any of them can serve as the per-partition candidate generator in front of our post-processing algorithms.

Filtered Vector Search. A growing line of work combines similarity search with predicate constraints over structured metadata [12]. Xie et al. [81] provide a unified framework for vector search with and without query predicates, analyzing the interplay between similarity search and predicate filtering. Graph-based filtered search methods include `ACORN` [59], `RWalks` [3], `DIGRA` [36], `iRangeGraph` [85], `DEG` [89], `Curator` [38], and `UNG` [7], and `SIEVE` [45] serves filtered queries from a collection of indexes built on subsets of the data. A recent in-depth experimental study [44] provides a comprehensive taxonomy of attribute filtering strategies in ANN search. `WoW` [77] introduces the first fully incremental index for range-filtered ANN with $O(\log^2 n)$ worst-case insertion complexity. `HONEYBEE` [97] addresses permission-aware vector search via dynamic partitioning, and timestamp-constrained ANN search has been studied in [75]. At SIGMOD and VLDB 2026, filtered search has become a topic of its own. `FAVOR` [65] performs filter-agnostic search using selectivity-aware exclusion distances, `PathSeer` [46] adapts neighbor handling to the filter during graph traversal, `CGIF` [52] combines proximity graphs with inverted files to support arbitrary predicates, Zhou et al. [99] balance efficiency and accuracy in filtered search, and Yang et al. [86] select among per-label indexes elastically for label-hybrid search. Several experimental studies and benchmarks appeared alongside: Lu et al. [51] study filter-agnostic vector search inside PostgreSQL, `VecBench` [94] and Lim et al. [48] provide controllable, hardness-aware benchmarks for filtered search, and Zhu et al. [100] and Wang et al. [73] evaluate hybrid vector–attribute querying. All of these methods enforce Boolean predicates on each returned item (e.g., "age $>$ 30"), whereas fairness-aware retrieval requires satisfying *count constraints* (e.g., "at least 2 and at most 4 items from group v") across multiple protected attributes simultaneously; none supports the per-group lower and upper count constraints that group fairness requires.

Group Fairness and Diversity. One of the initial works [67] treats multi-attribute group fairness independently per attribute. Islam et al. [31] enforce proportional representation in top-k results and show that the problem is weakly NP-hard for two attributes and strongly NP-hard for three attributes via 3-dimensional matching, directly motivating our ILP. Cai [6] generalizes

fair top-k selection to many protected groups: it searches for a linear scoring function, close to a reference one, whose top-k meets lower and upper bounds on the share of every group, where several attributes are combined into one composite attribute and intersectional groups may carry their own bounds. It shows that deciding whether a given scoring function is fair is already NP-hard in two dimensions when the number of groups is unbounded, and that a small k helps only when there are few groups. In our problem, the ranking (the distance to the query) is fixed, and the bounds apply to each of several attributes at once, so the fair answer is a selection of items rather than a choice of scoring function. Kurkure et al. [41] develop efficient algorithms for fair max-min diversification in $\mathbb{R}^d$, selecting k items subject to lower and upper bounds on items per group, the technically closest formulation to fairness-constrained k-NN among existing work at database venues. Mahabadi et al. [53] study the same lower/upper bound fairness constraint model in monotone submodular maximization subject to matroid constraints, achieving solutions satisfying fairness within a factor $(1-\varepsilon)$ in expectation. Cohen and Sternbach [13] apply facility location optimization to ensure fairness in query result retrieval, establishing NP-completeness for certain cases. Zhang et al. [93] study fair clique search ensuring balanced group representation in graphs, and Wang et al. [74] introduce fairness-aware community search on attributed graphs. On the theoretical side, Arenas et al. [4] study the tractability of diverse query answers, and Agarwal et al. [1] address well-representative summaries of conjunctive query results. Celis et al. [8] study classification with fairness constraints with provable guarantees. Oesterling et al. [57] propose multi-group proportional representation (MPR) as a fairness metric over intersectional groups, later extended to text-to-image generation [39]; MPR operates at the distributional level and does not certify per-attribute counts for every query. None of these approaches address approximate nearest neighbor search with multi-attribute count constraints over vector databases.

Mitigating Bias in Embedding Space. An orthogonal approach produces embeddings that mitigate bias via modified loss functions [91], adversarial training [18], or disentangled representations [14]. Post-processing methods such as CLIP-clip [70] and FairCLIP [72] address bias in vision models but cannot satisfy user-defined count constraints.

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